TrustFabric: ML-Based Claim Credibility Triage

19MAM56 Advanced Machine Learning Lab · CAT II Project Work

III M.Sc. AI & ML, V Semester · Coimbatore Institute of Technology

Experiment report · 2026-10-14

Abstract

This study builds a reproducible three-class claim-credibility classifier using the LIAR benchmark. The six original labels are aggregated into false, mixed, and true. A word-unigram TF-IDF/logistic-regression baseline is compared with a tuned word-and-character TF-IDF logistic-regression model. Official LIAR train, validation, and test partitions are preserved; a training-only three-fold GridSearchCV selects hyperparameters using macro-F1. The pipeline records class-wise and aggregate metrics, ROC/precision-recall curves, confusion matrices, coefficient inspection, and test-set errors. Its held-out results are 0.434 baseline macro-F1 and 0.445 improved-model macro-F1. Predictions reflect dataset labels and language patterns, not proof of truth. A separate FastAPI evidence-verification workflow and Docker packaging support local inference.

Index Terms— claim credibility, LIAR, text classification, TF-IDF, logistic regression, hyperparameter optimization, evidence-aware verification.

I. INTRODUCTION

Short factual statements are difficult to assess automatically because their meaning depends on context, source quality, time, and human annotation. This project treats classification as a review-prioritization task rather than an automated truth oracle. Its objectives are to document a real benchmark, prevent train/test leakage, compare a reproducible baseline with a tuned pipeline, analyse errors, and expose model output through a validated API.

II. LITERATURE REVIEW

Wang introduced LIAR as a benchmark of short political statements with six fact-checking labels [1]. The original paper's task differs from this three-class aggregation, so published scores are context rather than a direct comparator. Pérez-Rosas et al. studied automatic fake-news detection on a different dataset and task [3]; their reported results are likewise not numerically comparable here. TF-IDF and linear classifiers provide a transparent, reproducible text-classification baseline, implemented with scikit-learn [2].

III. DATASET AND PREPROCESSING

The source is the author-hosted LIAR archive [1]. Its README permits research use, states that original sources retain copyright, and provides no general redistribution license. The archive's 12,791 records comprise 10,240 train, 1,284 validation, and 1,267 test statements before cleaning. This experiment retains 12,765 unique usable statements; the official partitions remain separate. Blank statements, missing/unknown labels, and duplicate statement text are removed. Text is Unicode NFKC-normalized, control characters are stripped, and whitespace is collapsed. Six labels map to false (pants-fire/false), mixed (barely-true/half-true), and true (mostly-true/true).

Data limitations: LIAR contains short US political claims, reflects human judgments, and may be affected by source, speaker, and time biases. Three-class aggregation discards distinctions between original labels. Results do not establish current factual accuracy or generalization to other domains.

IV. METHODS AND EXPERIMENTAL DESIGN

The baseline uses word unigram TF-IDF (maximum 30,000 features) and class-weighted logistic regression. The improved estimator combines word n-gram TF-IDF (1–2 grams, up to 40,000 features) and character-boundary n-gram TF-IDF (3–5 grams, up to 20,000 features) before logistic regression. Both models use the same training, validation, and test partitions. The six-candidate grid searches word n-gram range and regularization strength $C \in \{0.25, 1, 4\}$; stratified three-fold CV is performed on training data only.

Validation results are measured on the original validation partition before refitting. After selection, baseline and improved estimators are refit on train plus validation and evaluated once on the held-out official test set. The random seed is 42. Vectorization resides within each scikit-learn pipeline, so each CV fold learns its vocabulary only from that fold's training data. The logged metrics are accuracy, macro precision/recall/F1, per-class precision/recall/F1/support, one-vs-rest macro ROC-AUC, macro average precision, and log loss. Because TF-IDF is sparse and already high-dimensional, dense PCA is not applied; capped word/character vocabularies and class-specific linear coefficients provide the representation analysis without discarding sparse lexical signals.

V. IMPLEMENTATION

The modular code separates data ingestion, model construction, training, visualization, and inference. `trustfabric.train` writes both joblib model artifacts, JSON benchmark and confusion matrices, CSV comparison and hyperparameter history, classification reports, and paired test predictions. Seven plots are generated from the actual LIAR distribution and held-out experiment: class counts, optimized and baseline confusion matrices, one-vs-rest ROC, precision–recall, model comparison, and coefficient-based feature inspection. The FastAPI model endpoint presents dataset-model output separately from evidence verification.

VI. RESULTS AND COMPARATIVE ANALYSIS

Model	Acc.	P	R	F1	ROC	PR
Baseline	0.436	0.434	0.438	0.434	0.616	0.435
Word+char tuned	0.448	0.446	0.450	0.445	0.631	0.450

Test-set baseline: Accuracy 0.436; macro precision 0.434; macro recall 0.438; macro-F1 0.434; macro one-vs-rest ROC-AUC 0.616; macro PR-AUC 0.435; log loss 1.067. Improved model: Accuracy 0.448; macro precision 0.446; macro recall 0.450; macro-F1 0.445; macro one-vs-rest ROC-AUC 0.631; macro PR-AUC 0.450; log loss 1.045. Best training-only CV macro-F1 was 0.431; selected parameters were {"classifier__C": 0.25, "features__word__ngram_range": [1, 2]}. Test support was 1,263 statements. The tuned model's gains are modest and are specific to this split and label mapping; no claim of state-of-the-art performance is made.

The most frequent error types and per-class results are saved in `outputs/metrics/error_analysis.csv` and `outputs/metrics/classification_report.json`. The comparison is reproduced under one shared test set; published experiments use incompatible data/label protocols and are not scored side-by-side.

VII. DEPLOYMENT AND LIMITATIONS

The FastAPI app serves a browser interface, `/api/predict`, `/api/v1/verify`, and health/metrics endpoints. The classifier endpoint returns a dataset-label prediction with an explicit non-fact-check warning. The separate evidence workflow searches for sources and compares passages with an NLI model; search snippets and NLI outputs are not verified ground truth. The Docker image pins direct dependencies, runs as a non-root user, and uses a model-aware health check. A public AWS endpoint is intentionally not asserted unless deployed and tested.

Limitations include subjective labels, political-domain bias, label aggregation, limited context, dataset age, possible distribution shift, and uncalibrated probabilities. The model should support human review, not make high-stakes decisions. Evidence retrieval depends on external search availability and can return incomplete, stale, or low-quality sources.

VIII. CONCLUSION AND FUTURE WORK

TrustFabric demonstrates a reproducible LIAR-based comparison with leakage-conscious preprocessing, a tuned TF-IDF ensemble of word and character representations, held-out analysis, a separate evidence verification workflow, and containerized inference. Future work should evaluate calibration, temporal and cross-domain robustness, additional annotators, transformer models under identical partitions, and human-in-the-loop decision support.

REFERENCES

[1] W. Y. Wang, “Liar, Liar Pants on Fire: A New Benchmark Dataset for Fake News Detection,” in *Proc. 55th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 2017, pp. 422–426, doi: 10.18653/v1/P17-2067.

[2] F. Pedregosa *et al.*, “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, no. 85, pp. 2825–2830, 2011.

[3] V. Pérez-Rosas, B. Kleinberg, A. Lefevre, and R. Mihalcea, “Automatic Detection of Fake News,” in *Proc. 27th International Conference on Computational Linguistics*, 2018, pp. 3391–3401.

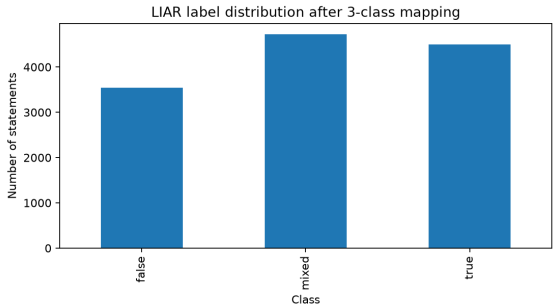


Fig. 1. Class counts after the three-way label mapping.

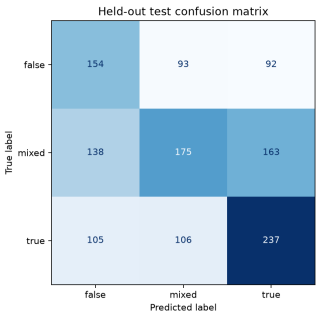


Fig. 2. Improved model confusion matrix on the held-out test set.

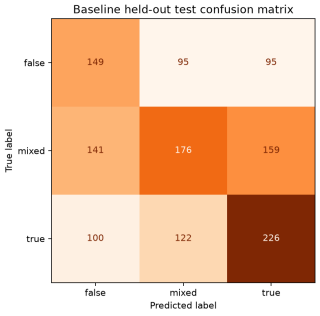


Fig. 3. Baseline confusion matrix on the same test set.

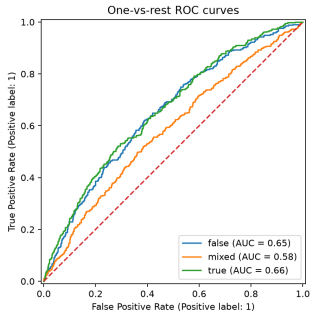


Fig. 4. One-vs-rest ROC curves for the improved model.

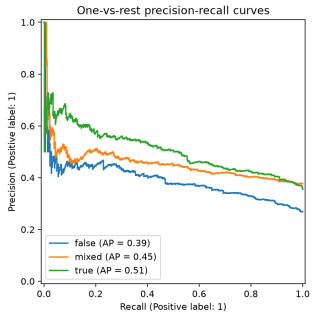


Fig. 5. One-vs-rest precision–recall curves.

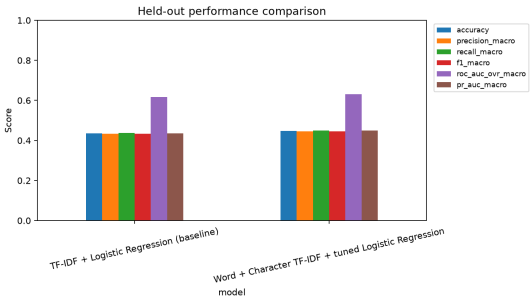


Fig. 6. Baseline/improved comparison on the shared test set.

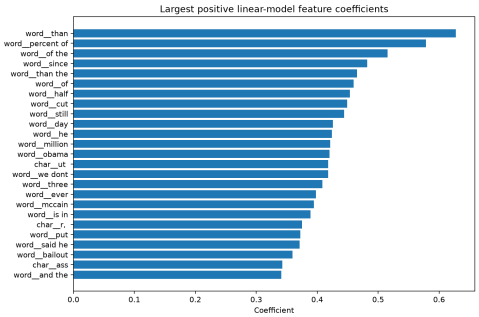


Fig. 7. Largest positive class coefficients from the linear model.